

Subject Name: Biochemical Engineering Laboratory

Subject No: BT29207

Prerequisite: Biochemical Reaction Engineering and Bioenergetics (BT20207)

Objective

This laboratory aims to provide students with hands-on experience in implementing some important concepts of Biochemical Engineering using Bioreactors/Fermenters, while initially learning about their constructional features and modes of operation. The course will teach them as to how to study death kinetics experimentally during the process of medium sterilization. Following this, the students will get to learn as to how to do cell culture in fermenter/reactor, study cell growth kinetics and quantify specific growth kinetic parameters. Further, the students will learn about various aspects and operational parameters that are critical for cell growth during bioreactor operation, for example determination of mixing time, volumetric mass transfer coefficient etc.. Finally, the students will be taught as to how to apply their knowledge in developing one or two products through cell culture experiments at laboratory scale. Thus this laboratory course will impart hands-on training and experience in important practical aspects of biochemical engineering including bioreactor operation that have critical importance in a Biotechnology industry.

L-T-P 0-0-3

Credit 2

Content:

1. To demonstrate various bioreactor configurations, parts and process control systems (3 hours)
2. To study in-situ sterilization of medium and air (3 hours)
3. To study microbial death kinetics: Determination of Nabla Factor and L90 Value (3 hours)
4. To study microbial cell growth kinetics in batch reactor and measure (a) maximum specific growth rate ($\mu_{\max}$), (b) saturation constant (K_S) and (c) growth yield coefficient ($Y_{X/S}$) (6 hours)
5. To measure mixing time in a fermenter and relate this determination to Reynolds number and Power number for specific set of impellers (3 hours)
6. To determine volumetric mass transfer coefficient (k_{La}) in a fermenter using dynamic gassing out method (6 hours)
7. To study lab-scale production of an important biotechnology product in a fermenter (6 hours)

8. To study indoor and outdoor cultivation of microalgae in closed and open photobioreactors (6 hours)

Text books:

Biochemical Engineering Fundamentals, by James E. Bailey and David F. Ollis, McGraw-Hill International Editions.

Bioprocess Engineering: Basic Concepts, Third Edition, by F. Kargi and Michael L. Shuler, Prentice Hall

Bioprocess Engineering Principles, Second Edition, by Paulin M Doran, Academy Press